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T11 – 05

Assignment No. 10

Aim: Design a feedback form using Flask.

LO Mapping: LO6: Design Web Applications using flask

Theory:

Flask is a lightweight and flexible **Python web framework** used for building web applications. It follows a **micro-framework** approach, meaning it provides the essential tools needed to build a web app without enforcing a strict structure or including unnecessary libraries by default. Flask is based on **Werkzeug** (a WSGI toolkit) and **Jinja2** (a templating engine). This makes it easy for beginners to start building web apps while still being powerful enough for more complex projects. Flask allows you to define routes, handle requests and responses, interact with databases, and render HTML templates, making it suitable for building both small and large-scale web applications.

Designing a web application in Flask typically involves setting up routes using Python functions, which are mapped to URLs. These functions can return HTML pages rendered with dynamic data using Jinja templates. Flask also supports form handling, session management, authentication, and database connectivity using extensions like **Flask-WTF**, **Flask-Login**, and **Flask-SQLAlchemy**. Developers can easily separate logic, templates, and static files (CSS, JS, images), which leads to clean and maintainable code. Flask's simplicity and flexibility make it a great choice for developing web apps like dashboards, blogs, portfolio sites, and even RESTful APIs.

Program:

```
<!-- APP.PY --> from flask import Flask, render_template, redirect, url_for
from flask_sqlalchemy import SQLAlchemy from flask_wtf import FlaskForm
from wtforms import StringField, TextAreaField, SelectField,
SubmitField from wtforms.validators import DataRequired, Email app =
Flask(__name__) app.config['SECRET_KEY'] =
'your_secret_key' app.config['SQLALCHEMY_DATABASE_URI'] =
'sqlite:///feedback.db' db = SQLAlchemy(app)
```

```

# Database Model class Feedback(db.Model):    id = db.Column(db.Integer,
primary_key=True)    name = db.Column(db.String(100), nullable=False)

email = db.Column(db.String(100), nullable=False)    service_rating =
db.Column(db.String(10),    nullable=False)    usability_rating =
db.Column(db.String(10), nullable=False)    comments = db.Column(db.Text,
nullable=True)


# Form using Flask-WTF class FeedbackForm(FlaskForm):    name =
StringField('Name', validators=[DataRequired()])    email =
StringField('Email', validators=[DataRequired(), Email()])    service_rating =
SelectField(    'Service Rating',    choices=[

        ('1', '1 - Poor'),

        ('2', '2 - Fair'),

        ('3', '3 - Good'),

        ('4', '4 - Very Good'),

        ('5', '5 - Excellent')

    ],    validators=[DataRequired()])

    usability_rating =
SelectField(

```

```

        'Usability Rating',      choices=[

            ('1', '1 - Difficult'),

            ('2', '2 - Fair'),

            ('3', '3 - Good'),

            ('4', '4 - Easy'),

            ('5', '5 - Very Easy')

        ],      validators=[DataRequired()]

    )    comments = TextAreaField('Additional

Comments')    submit = SubmitField('Submit')


@app.route('/', methods=['GET', 'POST']) def feedback():

    form = FeedbackForm()    if form.validate_on_submit():        new_feedback

= Feedback(            name=form.name.data,            email=form.email.data,

service_rating=form.service_rating.data,

usability_rating=form.usability_rating.data,

comments=form.comments.data

        )

    db.session.add(new_feedback)

```

```

        db.session.commit()    return redirect(url_for('thankYou'))    return

render_template('feedbackForm.html', form=form)

@app.route('/thankYou') def thank_you():

    return render_template('thankYou.html')

@app.route('/feedbackList') def feedback_list():

    feedbacks = Feedback.query.all()    return

render_template('feedbackList.html', feedbacks=feedbacks) if __name__ ==

'__main__':    with app.app_context():

        db.create_all()    app.run(debug=True)

<!-- THANK_YOU.HTML: -->

<!DOCTYPE html>

```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Thank You</title>
```

```
<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.m
in.css">

</head>

<body class="bg-light d-flex justify-content-center align-items-center vh-100">

<div class="card shadow p-4 text-center" style="width: 50%;">

<h2>Thank You!</h2>

<p>Your feedback has been submitted successfully.</p>

<a href="/" class="btn btn-success">Submit Another Feedback</a>

<a href="/feedback-list" class="btn btn-info">View Feedbacks</a>

</div>

</body>

</html>

<!-- FEEDBACK_LIST.HTML: -->

<!DOCTYPE html>

<html lang="en">
```

```
<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Feedback List</title> <link rel="stylesheet"

href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.m

in.css">

</head>

<body class="bg-light">
```



```
<div class="container mt-5">

<div class="card shadow p-4">

<h2 class="text-center mb-4">Submitted Feedbacks</h2>

<table class="table table-striped">

<thead>

<tr>

<th>Name</th>

<th>Email</th>

<th>Service Rating</th>

<th>Usability Rating</th>

<th>Comments</th>

</tr>

</thead>

<tbody>

{% for feedback in feedbacks %}

<tr>

<td>{{ feedback.name }}</td>

<td>{{ feedback.email }}</td>

<td>{{ feedback.service_rating }}</td>
```

```
<td>{{ feedback.usability_rating }}</td>
```

```
<td>{{ feedback.comments }}</td>
```

```
</tr>
```

```
{% endfor %}
```

```
</tbody>
```

```
</table>
```

```
<a href="/" class="btn btn-primary">Submit Another Feedback</a>

</div>

</div>

</body>

</html>

<!-- FEEDBACK_FORM.HTML: -->

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Feedback Form</title> <link rel="stylesheet"

href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.m

in.css">

</head>

<body class="bg-light">

<div class="container mt-5">

<div class="card shadow p-4">

<h2 class="text-center">Feedback Form</h2>
```

```
<form method="POST">

{{ form.hidden_tag() }}

<div class="mb-3">

{{ form.name.label(class="form-label") }}

{{ form.name(class="form-control") }}
```

```
</div>
```

```
<div class="mb-3">
```

```
{{ form.email.label(class="form-label") }}
```

```
{{ form.email(class="form-control") }}
```

```
</div>
```

```
<div class="mb-3">
```

```
{{ form.service_rating.label(class="form-label") }}
```

```
{{ form.service_rating(class="form-select") }}
```

```
</div>
```

```
<div class="mb-3">
```

```
{{ form.usability_rating.label(class="form-label") }}
```

```
{{ form.usability_rating(class="form-select") }}
```

```
</div>
```

```
<div class="mb-3">
```

```
{{ form.comments.label(class="form-label") }}
```

```
{{ form.comments(class="form-control", rows="3") }}
```

```
</div>
```

```
<div class="text-center">
```

```
{{ form.submit(class="btn btn-primary btn-lg") }}
```

```
</div>
```

```
</form>
```

```
</div>
```

```
</div>
```

```
</body>
```

```
</html>
```

Output:

Feedback Form

Name
Khalid Qureshi

Email
khalidqureshi1195@gmail.com

Service Rating
5 - Excellent

Usability Rating
4 - Easy

Additional Comments
No Comments

Submit